

IV CAL 2023

Esercizio 1

1) BIAS POINTS

HYPOTHESIS OF FORWARD ACTIVE ZONE FOR BOTH Q_1 AND Q_2

$$V_G = V_{CC} \cdot \frac{R_{G2}}{R_{G2} + R_{G1}} = 3,9 \text{ V} \quad R_G = R_{G1} \parallel R_{G2} = 18,22 \text{ K}\Omega$$

$$V_{GS} = V_G - I_D R_S$$

$$\begin{cases} I_D = \frac{V_G}{R_S} - \frac{V_{GS}}{R_S} \\ \frac{I_D}{I_{DSS}} = \frac{V_{GS}^2}{V_t^2} + 1 - \frac{2 V_{GS}}{V_t} \end{cases}$$

$$a = \frac{1}{V_t^2}$$

$$b = \frac{1}{I_{DSS} R_S} - \frac{2}{V_t}$$

$$c = 1 - \frac{V_G}{I_{DSS} R_S}$$

$$V_{GS_{12}} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \begin{matrix} \oplus & 3,75 \text{ V} \\ \ominus & -19,75 \text{ V} \end{matrix}$$

V_{GS} HAS TO BE GREATER THAN V_t AND SO

$$V_{GS} = 3,75 \text{ V}$$

$$I_D = 0,766 \text{ mA}$$

$$I_D = I_E$$

$$I_B = \frac{I_D}{\beta_F + 1} = 3,81 \mu A$$

$$I_C = \frac{I_E}{1 + \frac{1}{\beta_F}} = 0,762 mA$$

$$V_C = V_{CC} - I_C R_C = 5,75V$$

$$V_{eq} = V_{CC} \cdot \frac{R_{B2}}{R_{B1} + R_{B2}} = 5,62V$$

$$R_B = R_{B1} \parallel R_{B2} = 30,7 k\Omega$$

$$V_{DS} = V_{eq} - V_{BE} - I_B R_B - I_D R_S = 4,45V > V_{GS} - V_t = 1,75V$$

FAZ VERIFIED FOR Q_2

$$V_{CE} = V_C - V_{DS} - V_{I_D R_S} = 1,15V > V_{CE_{SAT}} \approx 0,2V$$

FAZ VERIFIED FOR Q_1

$$g_{m1} = \frac{I_C}{V_T} = 29,4 mS$$

$$r_{\pi} = 6,8 k\Omega$$

$$g_{m2} = \frac{2 I_D}{V_{GS} - V_t} = 0,875 mS$$

2) QUIESCENT OF POWER DISSIPATION

$$Q_1 \text{ DISSIPATES } P_1 = V_{CE} I_C + V_{BE} I_B = 0,879 \text{ mW}$$

$$Q_2 = V_{DS} \cdot I_D \quad \text{GLUOSTO}$$

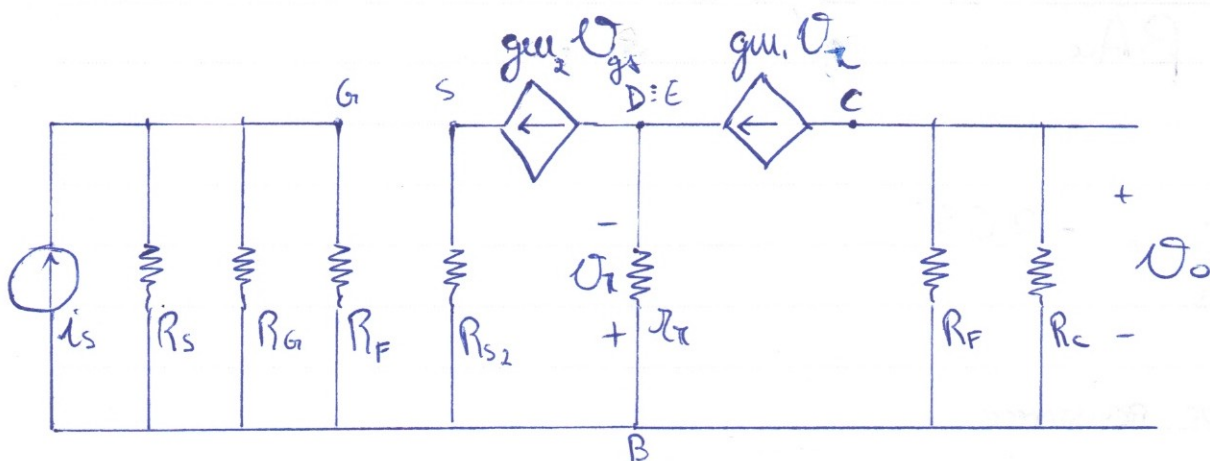
3) VOLTAGE GAIN

THE TOPOLOGY OF THE FEEDBACK IS CURRENT MIXING - VOLTAGE SENSING

THE β NETWORK IS MADE OF R_F

$$R_{if} = R_F \quad R_{of} = R_F \quad \beta = -1/R_F$$

THE OPEN LOOP SMALL SIGNAL EQUIVALENT CIRCUIT IS:



$$A_a = A_2 \cdot A_1 \cdot d_{in} = \frac{v_o}{v_e} \cdot \frac{v_e}{v_g} \cdot \frac{v_g}{i_s}$$

$$A_2 = \frac{V_o}{V_e} = g_{m1} R_F \parallel R_C = 212,1$$

$$\cdot V_o = -g_{m1} R_F \parallel R_C V_{\pi} = -g_{m1} R_F \parallel R_C (-V_e)$$

$$A_1 = \frac{V_e}{V_{in}} = \frac{-g_{m2} R_{in}^2}{1 + g_{m2} R_{s2}} = -0,0252$$

$$\cdot R_{in}^2 = \frac{r_{\pi}}{1 + \beta_0} = 34 \Omega$$

$$d_{in} = \frac{V_{in}}{i_s} = R_s \parallel R_F \parallel R_G = 5,53 \text{ k}\Omega$$

$$A_a = -31,156 \cdot 10^3$$

$$A_F = \frac{A_a}{1 + \beta A_a} = -20,5 \cdot 10^3$$

$$A_{VS} = \frac{A_F}{R_s} = -2,05$$

4) INPUT RESISTANCE

$$R_{in}^a = d_{in}$$

$$R_{in}^F = \frac{R_{in}^a}{1 + \beta A_a} = 3,84 \text{ k}\Omega$$

$$R_{in} = R_s + \frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_s}} = 16,23 \text{ k}\Omega$$

5) OUTPUT RESISTANCE

$$R_{out}^a = R_c \parallel R_F$$

$$R_{out}^F = \frac{R_{out}^a}{1 + \beta A_{u1}} = 4,75 \text{ k}\Omega$$

$$R_{out} = R_{out}^F$$

6) f_H CONSIDERING C_{cB}

$$R_{cB}^o = R_F \parallel R_c$$

$$\omega_H^a = \frac{1}{C_{cB} R_{cB}^o} = \cancel{438,62 \pi} \quad 6,3 \cdot 10^6 \frac{\text{RAD}}{\text{S}}$$

$$\omega_H^F = \omega_H^a (1 + \beta A_{u1}) = 9,57 \cdot 10^6 \frac{\text{RAD}}{\text{S}}$$

$$f_H = \frac{\omega_H^F}{2\pi} = 1,52 \cdot 10^6 \frac{\text{RAD}}{\text{S}}$$

ESERCIZIO 2

- 1) THE IDEAL CIRCUIT VOLTAGE GAIN IS THE ONE OF AN INVERTING CONFIGURATION AMPLIFIER

$$A_v = - \frac{R_2}{R_1} = - \frac{82}{33} = - 2,5$$

- 2) m_p AND B

$$\beta_o = \frac{R_1}{R_1 + R_2} = \frac{33}{33 + 82} = \frac{33}{115}$$

$$\frac{1}{\beta_o} = \frac{115}{33} = 3,48$$

$$A_1 = \frac{A_o \omega_o}{\omega_1} = \frac{10^5 \cdot 10^3}{2 \cdot 10^4} = 0,5 \cdot 10^3 = 500$$

$$\omega_{cr} = \sqrt{\frac{A_1}{1/\beta_o}} \quad \omega_1 = 239,6 \cdot 10^3 \frac{\text{RAD}}{\text{s}}$$

$$m_p = 180 - \text{AT}\left(\frac{\omega_{cr}}{\omega_o}\right) - \text{AT}\left(\frac{\omega_{cr}}{\omega_1}\right) = 4,8^\circ$$

A_f CAN BE APPROXIMATED AS $\frac{1}{\beta_o}$ FOR $\omega < \omega_{cr}$

So A_f STARTS TO GO DOWN WHEN $\omega = \omega_{CR}$.

THE BANDWIDTH CAN BE APPROXIMATED WITH ω_{CR} AS

3) THE COMPENSATING NETWORK INTRODUCES A POLE AND A ZERO

$$\omega_z = \frac{1}{C_c R_c} = 1,38 \text{ k} \frac{\text{RAD}}{\text{s}} > \omega_p = \frac{1}{C_c (R_c + R_1 // R_2)} = 1,93 \frac{\text{RAD}}{\text{s}}$$

$$\frac{1}{\beta_\infty} = \frac{R_1 // R_c + R_2}{R_1 // R_c} = 2,49 \cdot 10^3 > A_1$$

$$\omega'_{CR} = \frac{A_0 \omega_c}{1/\beta_\infty} = 4,023 \cdot 10^3 \frac{\text{RAD}}{\text{s}}$$

$$\begin{aligned} m_\varphi &= 180^\circ - AT\left(\frac{\omega'_{CR}}{\omega_c}\right) - AT\left(\frac{\omega'_{CR}}{\omega_c}\right) - AT\left(\frac{\omega'_{CR}}{\omega_p}\right) + AT\left(\frac{\omega'_{CR}}{\omega_z}\right) \\ &= 61,14^\circ \end{aligned}$$

FOR THE SAME REASON AS BEFORE, THE BANDWIDTH CAN STILL BE APPROXIMATED AS ω'_{CR}

4) IDEALLY $V^+ = V^-$ AND SO THE COMPENSATING
NETWORK DOESN'T PLAY ANY ROLE EFFECT.

$$A_N = - \frac{R_2}{R_1} = -2,5$$